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RAG chunking

Small-to-big Part 1 of 2

Parent document retriever

The top 4 chunks all missed the answer. Return the section around the top hit, and size that section yourself.

INSIDE

- How children map to parents
- A real run where the parent held the answer
- Four gotchas, with fixes
- When to use it

Cheat sheet at the end

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Match small, return the section

Small chunks give sharp embeddings but little context. LangChain's `ParentDocumentRetriever` (in `langchain-classic`) keeps both:

1. `parent_splitter` cuts each document into parents, such as sections.
2. `child_splitter` cuts each parent into small children.
3. Each child is embedded with its parent's id in `doc_id`.
4. Parents go into a docstore under that id.
5. A query finds the top `k` children, keeps their unique parent ids in rank order and returns those parents.

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The parent held the answer

A 4-section support handbook, 23 children of up to 100 characters. Query: "How long do warranty repairs take?"

Top 4 children (spaCy en_core_web_md vectors)

Chunk 1 0.918	The warranty covers manufacturing faults. It does not cover water damage or drops.
Chunk 2 0.905	Warranty claims need the serial number. Approved claims get a repair or a replacement.
Chunk 3 0.902	Damaged parcels are replaced at no cost when the customer sends a photo within 7 days.
Chunk 4 0.888	Refunds go back to the original payment method.

✕ The answer sentence ranked outside the top 4

Parent returned first: Warranty, 347 characters

Chunk 1	## Warranty … Approved claims get a repair or a replacement. **Repairs take about 10 business days.** …

✓ The answer comes back with the parent

Parent-document retriever

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No parent splitter: the whole document

`parent_splitter` defaults to `None`. Then the parent is the raw document you passed in. Same handbook, same 23 children:

`parent_splitter=None` default

✕ One parent per input document

1 parent, 1,360 characters: the whole handbook for every query

`500 characters, split on "\n## " first`

✓ One parent per section

4 parents: 401, 354, 347 and 251 characters, one per section

Load a 200-page PDF as one document and every hit returns all 200 pages. Split parents on headings first.

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`k` counts children

`search_kwargs` is empty by default, so the vector store's `k=4` applies to children. Parents are then deduped. The warranty query:

k=1 → Warranty only

1 parent, 347 characters

k=4 default → Warranty, Shipping, Refunds

3 parents, 1,102 characters

k=10 ✕ Every section of the handbook

4 parents, 1,353 characters

The query "Who has to approve refunds over $500?" returns 2 parents at k=4. Budget for k × your largest parent, then set `k` in `search_kwargs`.

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Re-ingest doubles every parent

`add_documents` gives each parent a new random id (`uuid4`). The same handbook, added twice:

```
children              23 → 46
parents                4 → 8
top 4 children        2 Warranty, each twice
returned              Warranty ×2 (two ids)
```

- The retriever has no delete method. Remove old children from the vector store and old parents from the docstore yourself.
- Passing `ids` with a `parent_splitter` fails unless there's one id per parent after splitting.

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In-memory parents vanish

With `InMemoryStore`, the docstore in the docs' example, parents live in the Python process. A new process, same vector store:

```
children in vector store   23
parents in docstore         0
retriever.invoke(query)    []
```

- No error: missing parents are filtered out and you get an empty list.
- Pass a persistent `byte_store`, such as `LocalFileStore`, or a database-backed store.

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Scores stay behind

Parents come back with the parent's own metadata: no similarity score and no record of which child matched.

```
parents[0].metadata
# {'source': 'handbook.md'}
```

- To cite the matching line, run `similarity_search_with_score` on the children yourself.
- The retriever's `similarity_score_threshold` mode drops weak hits, but needs relevance scores; `InMemoryVectorStore` raises `NotImplementedError`.

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LlamaIndex merges by ratio

LlamaIndex's `AutoMergingRetriever` swaps in a parent only when more than half of its children are retrieved.

Defaults in llama-index-core 0.14.25

```
AutoMergingRetriever(
    vector_retriever, storage_context,
    simple_ratio_thresh=0.5)
HierarchicalNodeParser.from_defaults()
# chunk_sizes=[2048, 512, 128]
```

- One matching child out of four comes back on its own.
- LangChain's retriever returns the parent on a single hit.

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Put it all together

Section-sized parents, small children, a set `k` and a docstore that survives a restart:

```
parent = RecursiveCharacterTextSplitter(
    chunk_size=2000,
    separators=["\n## ", "\n\n", "\n",
                ". ", " "])
child = RecursiveCharacterTextSplitter(
    chunk_size=400)
retriever = ParentDocumentRetriever(
    vectorstore=vectorstore,
    byte_store=LocalFileStore("./parents"),
    child_splitter=child,
    parent_splitter=parent,
    search_kwargs={"k": 4})
```

- LangChain's own example uses 2000 and 400 characters. Check k × 2000 characters against your token budget.
- On re-ingest, delete the old children and parents first.

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When it pays off

✓ Use it when

- Answers need the whole section: steps, conditions, exceptions
- Documents have headings to cut parents on
- k × your largest parent fits the context budget

→ Move on when

- Sections are long and the answer sits in a sentence or two: try sentence-window retrieval
- Sections run to thousands of tokens
- You need scores or the matched line for citations

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RAG chunking **Parent-document retriever**

Cheat sheet

- [x] Embed small children, return their parent section
- [x] `parent_splitter=None` returns the whole document
- [x] Default `k=4` counts children; parents deduped
- [x] Budget k × your largest parent
- [x] Re-adding duplicates: delete old copies first
- [x] Persist the docstore, or a restart returns []
- [x] Parents come back without scores

From the RAG series by Shahbaz Khalid

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Tomorrow →

RAG chunking

Small-to-big ■□ Part 2 of 2

Sentence window retrieval

Go smaller still: embed single sentences, then hand the model the sentences around each hit.

✓ **Today: Parent-document retriever**